Thesis
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Data

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Empirical Model

The model includes the first job since labor market entrance for analysis. Let $t = 0, \ldots, T_i^1$ denotes observations when worker i remains in their first job. The sample ends when the worker leaves the first job.

Let x_{itc}^1 denote the cumulate tenure in the first job for worker i from cohort $c \in \{79, 97\}$measured in years, with $x_{itc}^1 = 0$ at the beginning of that job.

Let $J_{ic}^1 \in \{1, \ldots, J\}$ denote the occupation in which worker i begins the first job, and define $D_{ijc}^1 = \mathbb{1}\{J_{ic}^1 = j\}$ which is fixed at the occupation observed when the first job begins.

Baseline Theoretical Model

Following Altonji and Pierret (2001), worker productivity depends on characteristics that differ in their observability to employers and the econometrician. Adapted to the first job constraint, let log productivity be

$$p_{itc} = r_c S_{ic} + \alpha_c q_{ic} + \Lambda_c Z_{ic} + \eta_{ic} + H_c(x_{itc}^1)\,, \tag{1}$$

where S_{ic} is the education when the worker's first job begins; q_{ic} is information observed by employers but not fully observed by the econometrician, such as interview performance or hiring information; Z_{ic} is productivity related characteristic, which is observed by econometrician but not directly observed by employers at labor market entry, such as ASVAB measure; η_{ic} is productivity that initially neither observed by employer or econometrician; and $H_c(x_{itc}^1)$ represents the general productivity and wage experience profile during the first job.

At the beginning of the first job, the employer's information set is $\mathcal{I}_{i0c} = \{S_{ic}, q_{ic}, X_{i0c}\}$. The employer does not initially observe Z_{ic}, η_{ic}, or the true productivity. During the first

job, the employer observes noisy performance signals

$$y_{i\tau c} = p_{ic} + \nu_{itc}\,, \text{where } \mathrm{E}[\nu_{itc}] = 0\,, \mathrm{Var}(\nu_{itc}) = \sigma^2_{\nu c}\,,$$

for $\tau = 0, \ldots, t-1$. Then the information set could be expand as

$$\mathcal{I}_{itc} = \mathcal{I}_{i0c} \cup \{y_{i0c}, \ldots, y_{i,t-1,c}\}\,. \tag{2}$$

Under the competitive wage assumption, the worker's wage equals the expected productivity conditional on the information available at time t, then

$$\ln w_{itc} = \mathrm{E}[p_{itc} \mid \mathcal{I}_{itc}]\,.$$

The posterior expectation can be represented as

$$\mathrm{E}[p_{itc} \mid \mathcal{I}_{itc}] = \left(1 - \theta_c(x^1_{itc})\right) \mathrm{E}[p_{ic} \mid S_{ic}, q_{ic}] + \theta_c(x^1_{itc})\bar{y}_{itc} + H_c(x^1_{itc})\,, \tag{3}$$

where $\bar{y}_{itc}$ represents the performance signals observed to date and $\frac{\partial \theta_c(x)}{\partial x} > 0$. Thus, employers place less weight on initial signals and more weight on observed performance as tenure increase.

Baseline empirical specification

For cohort c, we estimate

$$\begin{aligned}\ln w_{itc} =& \alpha_c + \beta_{Sc} S_{ic} + \beta_{Zc} Z_{ic} + \gamma_{Sc}(S_{ic} \times x^1_{itc}) + \gamma_{Zc}(Z_{ic} \times x^1_{itc}) \\ &+ \theta_{1c} x^1_{itc} + \theta_{2c}(x^1_{itc})^2 + X'_{itc}\delta_c + \lambda_{tc} + \varepsilon_{itc}\,,\end{aligned} \tag{4}$$

where X_{itc} contains demographic and labor market controls, λ_{tc} denotes year effects, and ε_{itc} represents the wage residual.

The relevant coefficients are

$$\gamma_{Zc} = \frac{\partial^2 \mathrm{E}[\ln w_{itc}]}{\partial Z_{ic} \partial x^1_{itc}} \quad \text{and} \quad \gamma_{Sc} = \frac{\partial^2 \mathrm{E}[\ln w_{itc}]}{\partial S_{ic} \partial x^1_{itc}}\,.$$

The baseline model employer learning predictions are $H^{EL}_1 : \gamma_{Zc} > 0$ and, under statistical discrimination based on schooling, $H^{SD}_1 : \gamma_{Sc} < 0$.

The model would be estimated separately for the NLSY79 and NLSY97 cohorts, where the coefficients could differ across cohorts.

Adding first job occupation factors

Motivated from Altonji (2005), which extends employer learning by emphasizing that jobs differ in how strongly productivity depends on worker skill and how informative job performance is about that skill. The model predicts more rapid learning in jobs with greater skill requirements, while assignment to less skill-intensive jobs may reveal relatively little about underlying talent. The paper therefore provides a basis for allowing both skill prices and signal precision to vary across the worker's initial occupation.

Gibbons et al. (2005) model sectors as having different production technologies and different returns to observed and unobserved worker skills. In their model, sector-specific coefficients determine how strongly each sector values worker characteristics. Learning about initially unobserved ability then affects wages and sector allocation over time. However, as the sample is restricted to the first job, the mobility component of Gibbons et al. (2005) is not estimated.

Occupation extended theoretical model

Let the first job productivity in occupation j be

$$p_{ijtc} = a_{jc} + r_{S,jc} S_{ic} + \alpha_{jc} q_{ic} + \Lambda_{jc} Z_{ic} + \eta_{ijc} + H_{jc}(x^1_{itc}) \,. \tag{5}$$

Compared with Equation (1), $a_{jc}, r_{S,jc}, \Lambda_{jc}, H_{jc}(x)$ are allowed to vary by first job occupation.

Employers in occupation j observe a sequence of signals $y_{ij\tau c} = p_{ijc} + \nu_{ij\tau c}$, where $\mathrm{E}[\nu_{ij\tau c} \mid j, c] = 0$ and $\mathrm{Var}(\nu_{ij\tau c} \mid j, c) = \sigma^2_{\nu,jc}$. A smaller value of $\sigma^2_{\nu,jc}$ means that performance is more informative in occupation j, where employers learn more rapidly.

Let the initial uncertainty about productivity in occupation j be $\sigma^2_{0,jc}$. Assuming signals are independently and normally distributed, the posterior weight on accumulated performance information after x periods can be written as

$$\theta_{jc}(x) = \frac{x\sigma^2_{0jc}}{\sigma^2_{\nu jc} + x\sigma^2_{0jc}} \,. \tag{6}$$

Define the one period learning parameter to be

$$k_{jc} = \frac{\sigma^2_{0jc}}{\sigma^2_{0jc} + \sigma^2_{\nu jc}},$$

we can rewrite Equation (6) as

$$\theta_{jc}(x) = \frac{x k_{jc}}{1 + (x-1)k_{jc}}.$$

The employer's posterior expectation is therefore

$$\mathrm{E}[p_{ijtc} \mid \mathcal{I}_{ijtc}] = \left(1 - \theta_{jc}(x^1_{itc})\right) \mathrm{E}[p_{ijc} \mid S_{ic}, q_{ic}] + \theta_{jc}(x^1_{itc})\bar{y}_{ijtc} + H_{jc}(x^1_{itc})$$

Empirical specification with occupational variation

Adding occupational variations to the empirical model, we can now rewrite Equation (4) as

$$\begin{aligned}\ln w_{itc} =& \alpha_c + \beta_{Sc} S_{ic} + \beta_{Zc} Z_{ic} + \gamma_{Sc}(S_{ic} x^1_{itc}) + \gamma_{Zc}(Z_{ic} x^1_{itc}) + \theta_{1c} x^1_{itc} + \theta_{2c}(x^1_{itc})^2 \\ &+ \sum_{j=2}^{J} D^1_{ijt} \left[\delta_{jc} + \rho_{Sjc} S_{ic} + \rho_{Zjc} Z_{ic} + \kappa_{1jc} x^1_{itc} + \kappa_{2jc}(x^1_{itc})^2 + \phi_{Sjc}(S_{ic} x^1_{itc}) + \phi_{Zjc}(Z_{ic} x^1_{itc})\right] \\ &+ X'_{itc}\delta_c + \lambda_{tc} + \varepsilon_{itc}, \end{aligned} \tag{7}$$

For the omitted reference occupation $j = 1$, the ability based employer learning coefficient is $EL^Z_{1c} = \gamma_{Zc}$. For occupation $j \geq 2$, $EL^Z_{jc} = \gamma_{Zc} + \phi_{Zjc}$, where ϕ_{Zjc} measures how the rate at which wages become more strongly related to ability in occupation j differs from the rate in the reference occupation. Similarly, employer learning on the schooling coefficient for the reference occupation and occupation j is $EL^S_{1c} = \gamma_{Sc}$ and $EL^S_{jc} = \gamma_{Sc} + \phi_{Sjc}$, respectively.

Tests on whether the employer learning patter differs across occupations would be

$$H_0 : \phi_{Z2c} = \cdots = \phi_{ZJc} = 0 \text{ and } H_0 : \phi_{S2c} = \cdots = \phi_{SJc} = 0.$$

References

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